

Project 1 - Home SOC Lab Build Guide

Project 1 — Home SOC Lab

How to Use This Guide

PROJECT 1 — Home SOC Lab

- 1.1 Hardware Requirements
- 1.2 Choose Your Hypervisor
- 1.3 Network Topology
- 1.4 VM Build Order
- 1.5 Critical Configuration — Logging
- 1.6 Snapshot Strategy
- 1.7 Validation Checklist
- 1.8 Common Pitfalls

Project 1 — Home SOC Lab

**Standalone Build Guide — Extracted from the SOC Analyst Projects
Detailed Build Guide**

PROJECT 1 — Home SOC Lab

Goal: Build a self-contained network where you can run attacks, generate telemetry, and practice detection without breaking anything real.

Prerequisites: None. This is your foundation.

Time budget: 15–25 hours over 1–2 weeks.

1.1 Hardware Requirements

Minimum host machine to run this comfortably:

Resource	Minimum	Recommended
RAM	16 GB	32 GB
CPU	4 cores with VT-x/AMD-V	6+ cores
Storage	250 GB SSD	500 GB+ NVMe
OS	Windows 10/11 Pro, macOS, or Linux	Same

If you only have 16 GB RAM, you'll run 2–3 VMs at a time, not all of them. That's fine — start/stop as needed.

If you have less than 16 GB or older hardware, use **free Azure credits** (\$200 first month) or **Oracle Cloud's Always Free tier** for parts of this lab.

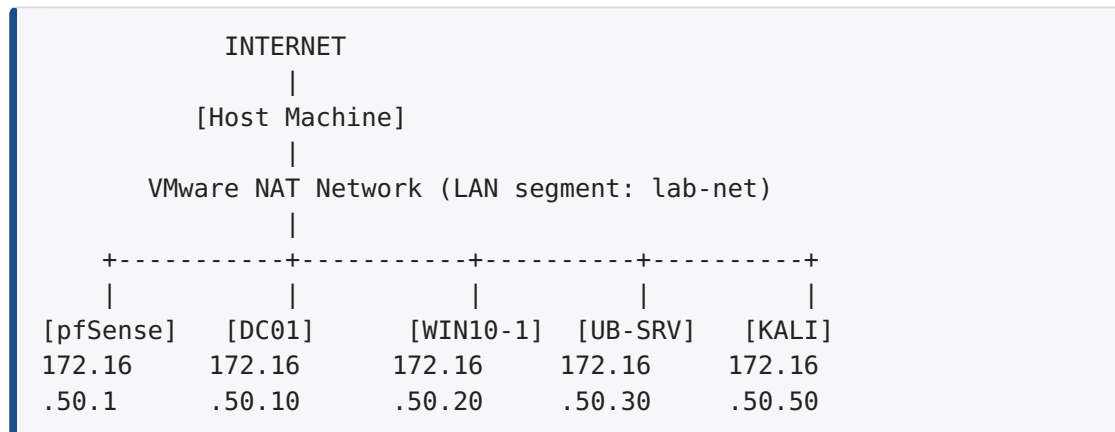
1.2 Choose Your Hypervisor

Hypervisor	When to pick it
VMware Workstation Pro (free as of 2024)	Best UX on Windows/Linux
VMware Fusion (free for personal use)	macOS
VirtualBox	Backup option, slower networking
Proxmox VE	If you have spare physical hardware to dedicate
Hyper-V	Windows 11 Pro built-in; works but limited networking

Recommendation for a learner: **VMware Workstation Pro** on a Windows host. Pick it and stop deliberating.

1.3 Network Topology

Plan this BEFORE you create VMs. Draw it on paper.



Why this subnet? We deliberately avoid `192.168.0.0/24` and `192.168.1.0/24` because nearly every home router (TP-Link, D-Link, ASUS, most ISP-issued routers) lives at `192.168.1.1`. If we put the lab there, every time you opened `https://192.168.1.1` to manage pfSense, you'd hit your real router's admin page instead. We also avoid `192.168.0.x` (also a common home range), `10.0.0.x` (used by some routers and most corporate VPNs), and `172.17.0.0/16` (Docker's default). The `172.16.50.0/24` range is RFC 1918 private space, supported by all firewalls, and almost never used by consumer equipment — so the lab can't accidentally collide with anything else on your home network.

In VMware: create a custom network (Edit → Virtual Network Editor → Add Network → Host-only or NAT). Call it `lab-net`. Disable DHCP on that segment (pfSense will handle it). When VMware asks for the subnet, set it to `172.16.50.0` with mask `255.255.255.0`.

1.4 VM Build Order

Build in this exact order. Each one depends on the previous.

VM 1 — pfSense Firewall (Gateway)

Why first: Everything else lives behind this. It segments your lab from the internet and from your host.

Steps:

1. Download pfSense Community Edition ISO from netgate.com
2. Create VM: 2 vCPU, 2 GB RAM, 20 GB disk, **2 network adapters**:
 - Adapter 1 → NAT (this is the WAN, gets internet)
 - Adapter 2 → `lab-net` (this is the LAN)
3. Boot from ISO, accept defaults, install
4. After reboot:
 - Assign interfaces: WAN = em0, LAN = em1
 - Set LAN IP to 172.16.50.1/24
 - Enable DHCP on LAN: range 172.16.50.100–200 (we'll set static IPs below this range)
5. From any other VM later, browse to `https://172.16.50.1` → admin / pfsense → change password immediately
6. Enable **pfSense logging to remote syslog** (you'll point this at Wazuh in Project 2)

Validation: From inside pfSense web UI, you can see the LAN interface up. Don't add other VMs yet.

VM 2 — Windows Server 2022 (Domain Controller)

Why next: The DC produces the richest, most valuable telemetry in your lab — and it's what every real corporate SOC monitors.

Steps:

1. Download Windows Server 2022 Evaluation ISO from microsoft.com (180-day trial, can be rearmed)
2. Create VM: 2 vCPU, 4 GB RAM, 60 GB disk, 1 NIC on `lab-net`
3. Install Windows Server Standard (Desktop Experience)
4. Set static IP: 172.16.50.10, mask 255.255.255.0, gateway 172.16.50.1, DNS 127.0.0.1
5. Rename computer to `DC01`, reboot
6. Server Manager → Add Roles and Features → **Active Directory Domain Services + DNS**
7. Promote to Domain Controller: create new forest, domain name `corp.local`, set DSRM password
8. After reboot, log in as `CORP\Administrator`
9. Create test users (Active Directory Users and Computers):
 - `jsmith` (regular user) — password `Password123!`
 - `aadmin` (Domain Admin) — password `AdminPass456!`
 - `svc_backup` (service account)

- `msscott` (regular user)

10. Create an OU structure: `Users`, `Workstations`, `Servers`, `ServiceAccounts`

Validation:

- `Get-ADUser -Filter *` in PowerShell shows your users
- `nslookup dc01.corp.local` from the DC returns its own IP

VM 3 — Windows 10/11 Client

Steps:

1. Download Windows 10/11 Enterprise Evaluation ISO
2. Create VM: 2 vCPU, 4 GB RAM, 60 GB disk, NIC on `lab-net`
3. Set static IP: 172.16.50.20, DNS 172.16.50.10 (your DC)
4. Rename to `WIN10-1`
5. Join domain `corp.local`:

```
Add-Computer -DomainName corp.local -Credential (Get-Credential) -  
Restart
```

6. Log in as `corp\jsmith` after reboot

Validation:

- `whoami /upn` shows `jsmith@corp.local`
- `gpupdate /force` succeeds

VM 4 — Ubuntu Server

Steps:

1. Ubuntu Server 22.04 LTS or 24.04 LTS ISO
2. VM: 2 vCPU, 2 GB RAM, 40 GB disk, NIC on `lab-net`
3. Static IP: 172.16.50.30, gateway 172.16.50.1, DNS 172.16.50.10
4. Install OpenSSH server during setup
5. After install: `sudo apt update && sudo apt upgrade -y`
6. Install useful tools:
`sudo apt install -y net-tools tcpdump htop curl wget unzip`

This VM will later host Wazuh in Project 2.

VM 5 — Kali Linux (Attacker)

Steps:

1. Download Kali pre-built VMware image from kali.org
2. Import to VMware, place NIC on `lab-net`
3. Static IP: 172.16.50.50
4. Default creds: kali / kali — change them
5. Update: `sudo apt update && sudo apt full-upgrade -y`
6. Confirm key tools work: nmap, metasploit, hydra, nikto, mimikatz (via Impacket)

Validation: From Kali, run `nmap -sV 172.16.50.10` — you should see open ports on the DC (53, 88, 389, 445, 3389, etc.)

1.5 Critical Configuration — Logging

This is the part 90% of home labs get wrong. **Without logs, this is just a network, not a SOC lab.**

A. Install Sysmon on every Windows VM

Sysmon is Microsoft's deep telemetry agent — without it, native Windows logs miss most attack behaviors.

```
# On each Windows VM (DC01, WIN10-1):
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip" -OutFile sysmon.zip
Expand-Archive sysmon.zip -DestinationPath C:\Sysmon

# Use the SwiftOnSecurity config – battle-tested, sane defaults
Invoke-WebRequest -Uri "https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-config/master/sysmonconfig-export.xml" -
OutFile C:\Sysmon\sysmonconfig.xml

C:\Sysmon\Sysmon64.exe -accepteula -i C:\Sysmon\sysmonconfig.xml
```

Validate: Event Viewer → Applications and Services Logs → Microsoft → Windows → **Sysmon** → Operational should be populating.

B. Enable PowerShell Logging

Run on each Windows VM in elevated PowerShell:

```

# Module logging
New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ModuleLogging" -Force
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ModuleLogging" -Name "EnableModuleLogging" -Value 1
New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ModuleLogging\ModuleNames" -Force
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ModuleLogging\ModuleNames" -Name "*" -Value "*"

# Script Block logging – captures the actual code that ran
New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Force
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Name "EnableScriptBlockLogging" -Value 1

```

C. Enable Command-line auditing

Group Policy on DC01:

- Computer Configuration → Administrative Templates → System → Audit Process Creation → **Enabled**
- Computer Configuration → Administrative Templates → System → Audit Process Creation → **"Include command line in process creation events"** → **Enabled**

Then `gpupdate /force` on all Windows VMs.

This makes Event ID 4688 include the full command line of every process — gold for SOC analysis.

D. Enable advanced audit policies on DC01

```

auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"Logoff" /success:enable
auditpol /set /subcategory:"Account Lockout" /success:enable /failure:enable
auditpol /set /subcategory:"Process Creation" /success:enable
auditpol /set /subcategory:"Process Termination" /success:enable
auditpol /set /subcategory:"Kerberos Service Ticket Operations" /success:enable /failure:enable
auditpol /set /subcategory:"Kerberos Authentication Service" /success:enable /failure:enable
auditpol /set /subcategory:"Credential Validation" /success:enable /failure:enable

```


1.6 Snapshot Strategy

After everything works, **take a clean snapshot of every VM**, named `clean-baseline-2026-MM-DD`. You will break things. Snapshots let you revert instantly.

Recommended snapshots:

- `clean-baseline` (after build)
 - `pre-sysmon` (delete after Sysmon proven stable)
 - `pre-attack` (before each Project 5 hunt)
-

1.7 Validation Checklist

Tick all before moving on:

- ☐ All 5 VMs can ping pfSense and each other
 - ☐ All VMs have internet through pfSense
 - ☐ WIN10-1 successfully joined to corp.local
 - ☐ Sysmon Operational log is producing events on every Windows host
 - ☐ Event ID 4688 in Security log shows full command lines
 - ☐ PowerShell Operational log captures script blocks
 - ☐ You can SSH from Kali to UB-SRV
 - ☐ You can RDP from your host to WIN10-1 (via pfSense port forward if needed) — *optional*
 - ☐ You took clean snapshots
-

1.8 Common Pitfalls

- **pfSense LAN unreachable:** forgot to disable VMware's NAT DHCP. Turn off DHCP on the host-only/NAT network in Virtual Network Editor.
- **Domain join fails:** client DNS pointing at 8.8.8.8 instead of the DC. Always point domain-joined clients at the DC for DNS.

- **Sysmon spamming logs:** That's the point. If your event log fills, increase max log size: Event Viewer → right-click log → Properties → Maximum log size = 512 MB.
 - **Out of RAM:** shut down VMs you're not actively using. Only DC01 + the one client + Kali during attack scenarios.
-